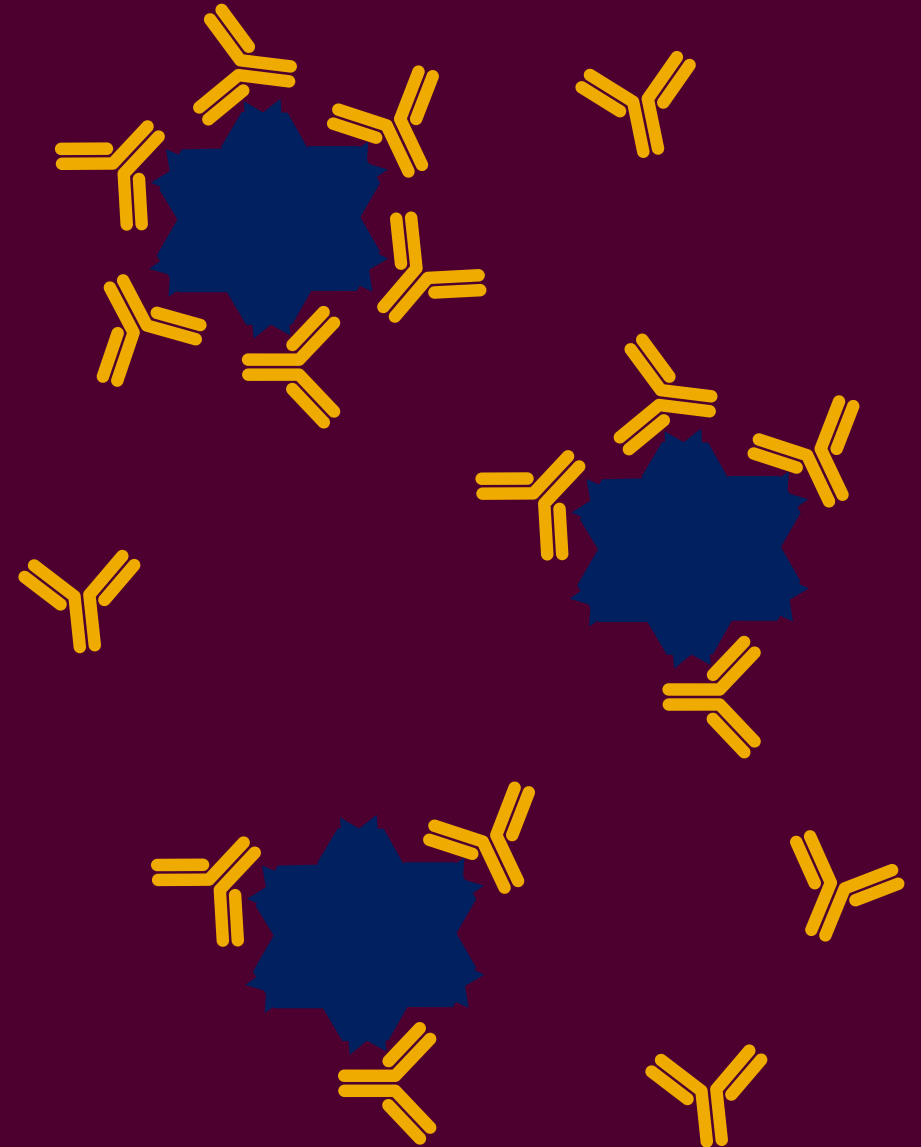




Sponsorship Spotlight: Continuous Counter- Current Affinity Nanoparticle Dialysis (C³ANDo)

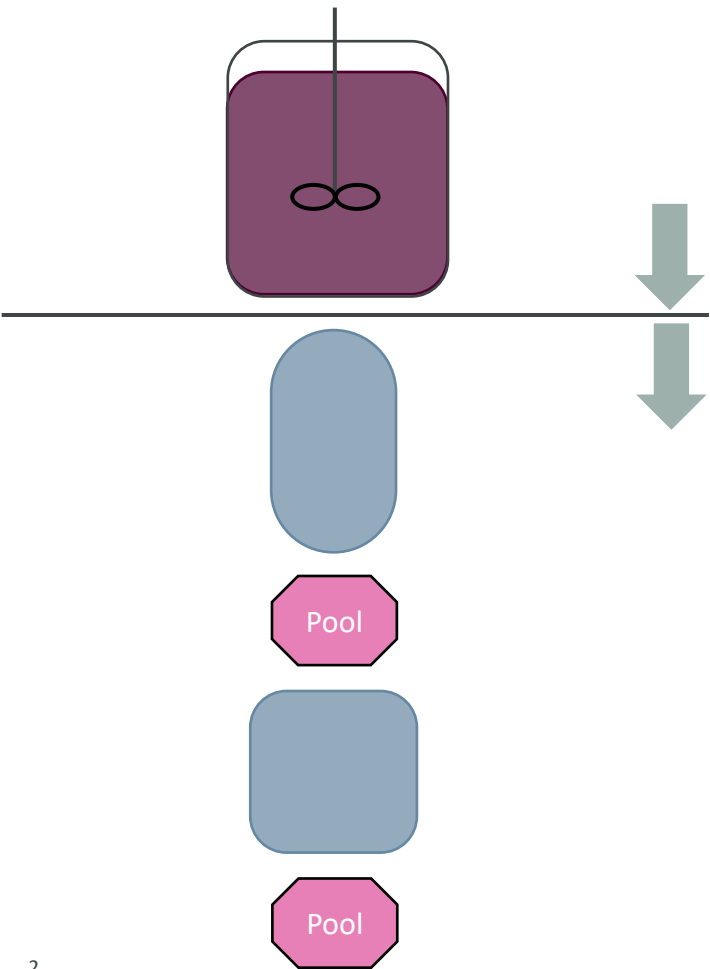
Taylor Bloom, Melissa Cyr, Ujwal Patil,
Kareem Fakhfakh, Joel Graham, Irina
Ramos, Jon Coffman

13 April 2025

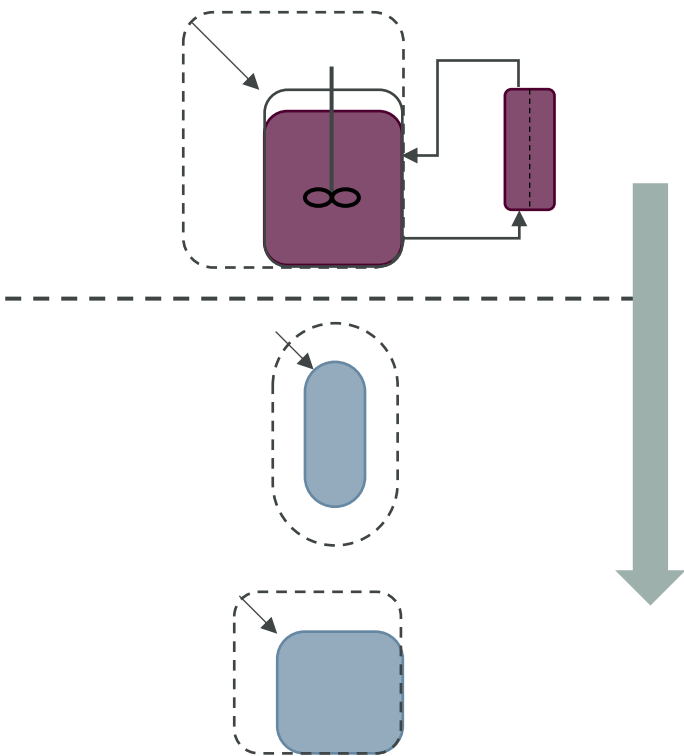


NGM Implementation in Two Stages: “3X” and “5X”

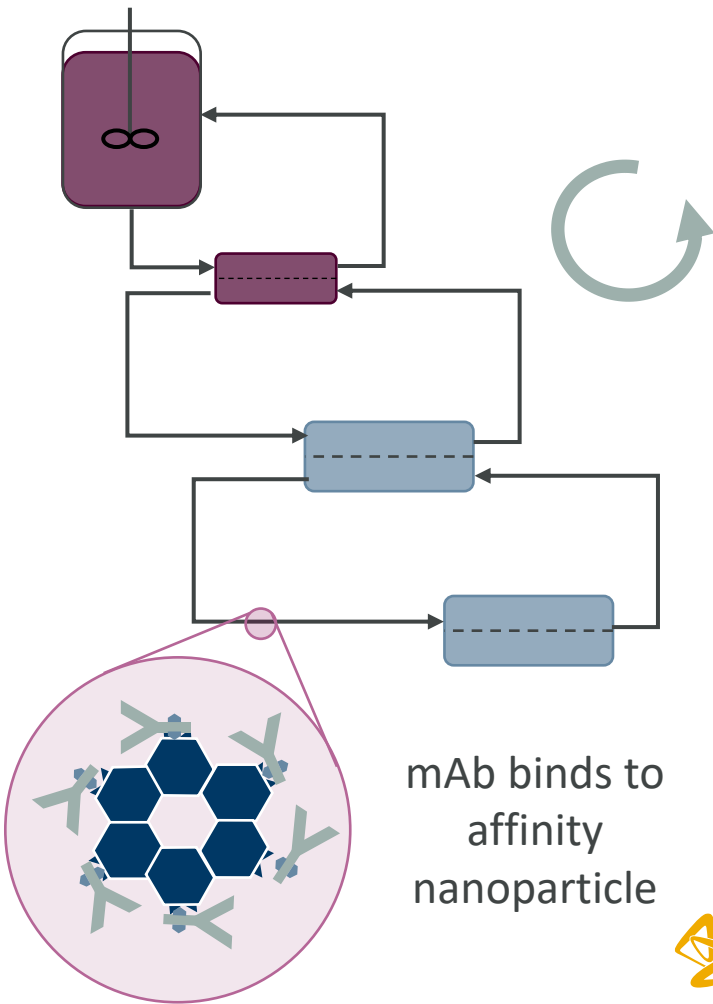
Current Manufacturing



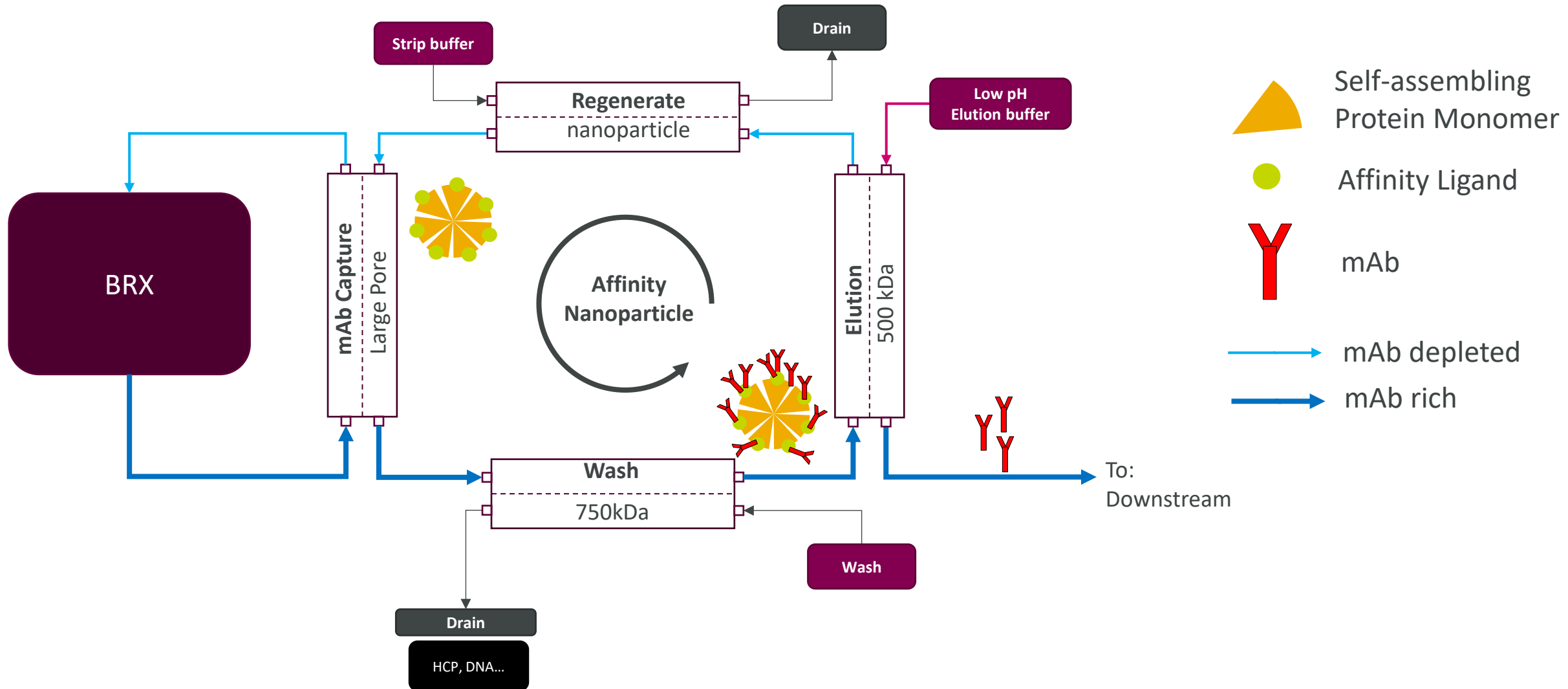
“3x2021” Manufacturing



“5x2025” Manufacturing

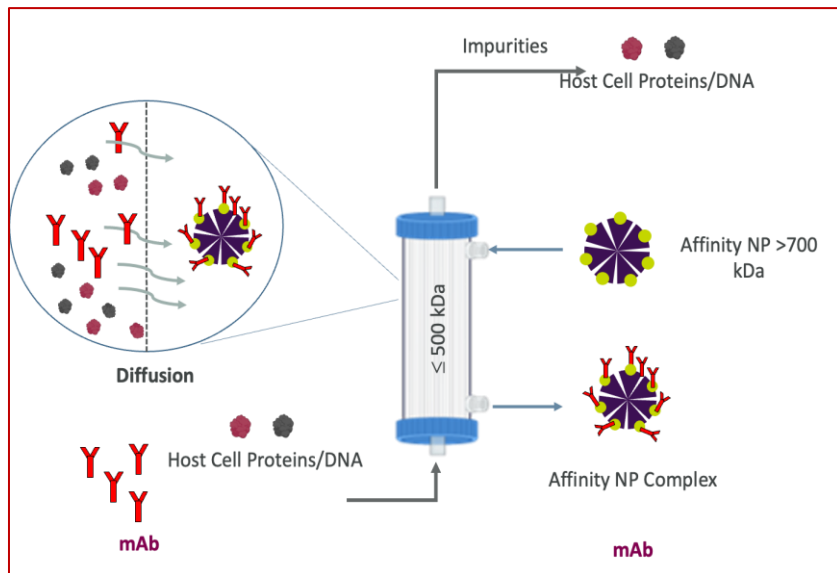


Affinity Nanoparticle Filtration Cycle

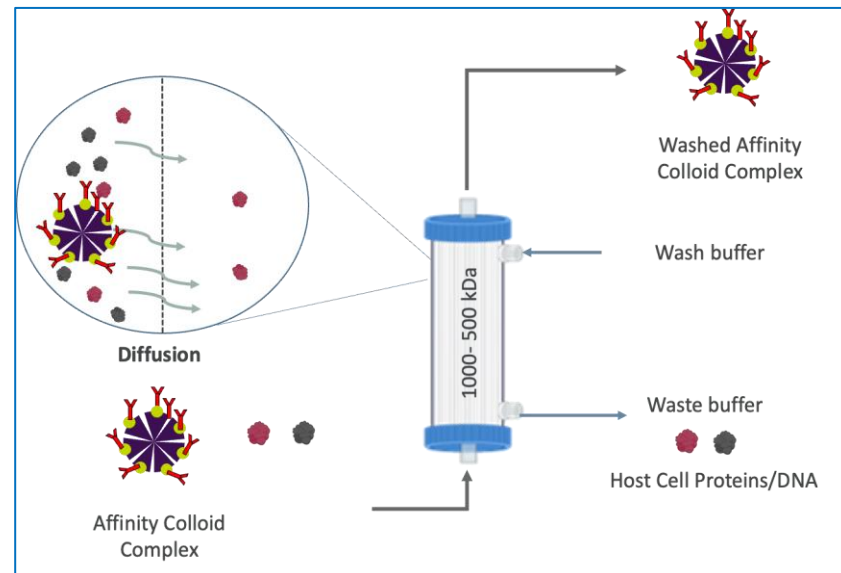


Continuous Counter-Current Affinity Nanoparticle Dialysis (C³ANDo)

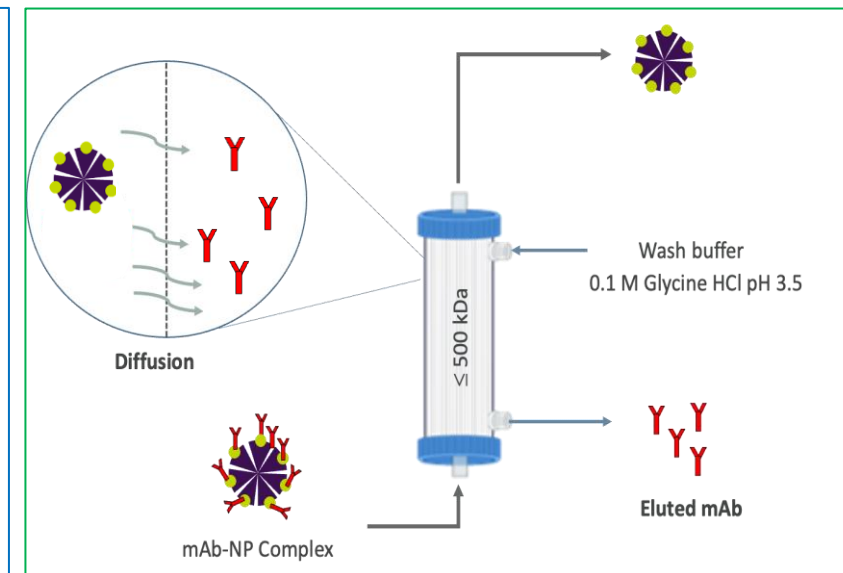
Capture



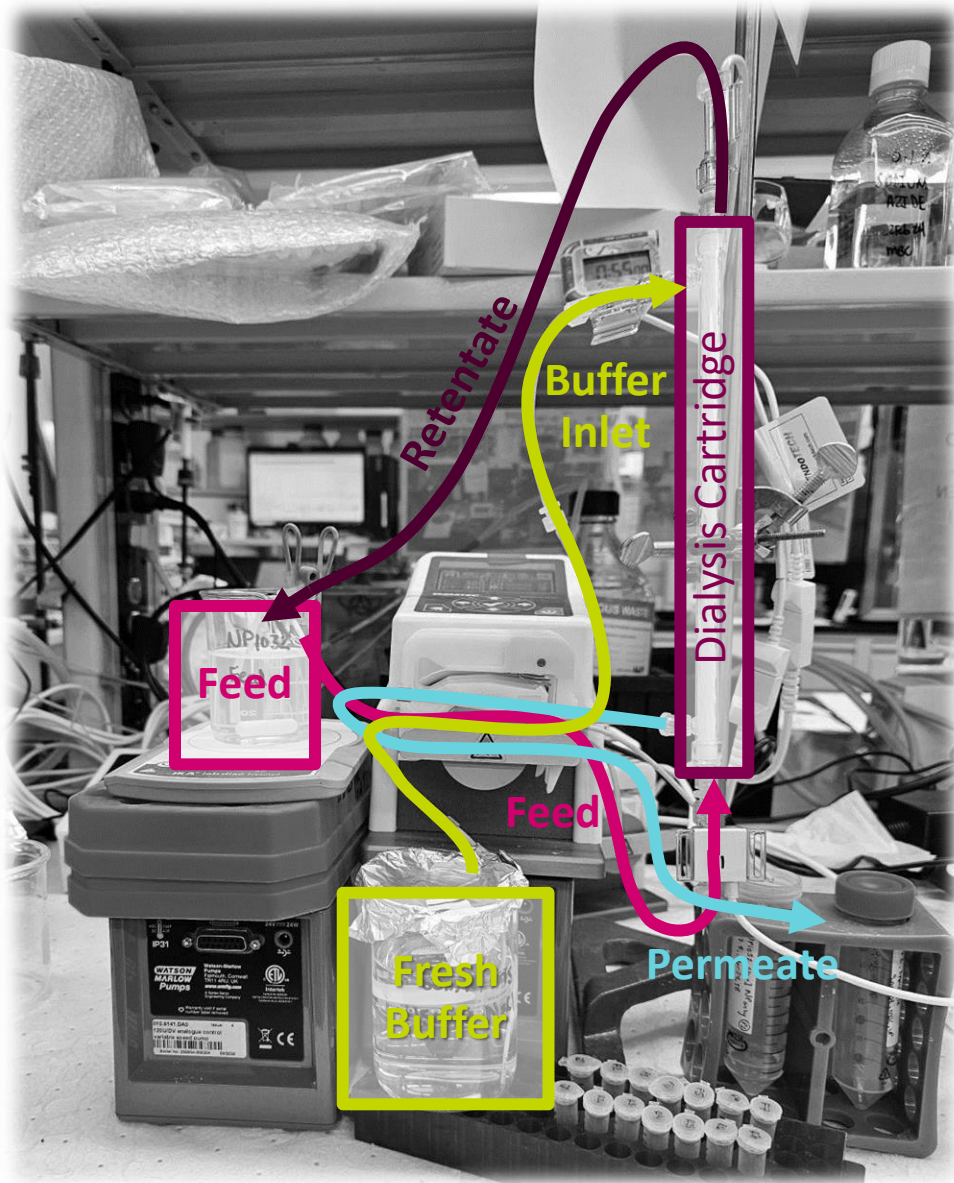
Wash



Elution



Modeling Antibody Removal



- Perform experiments to determine diffusivity values
- Use mathematical models to determine protein removal at full-scale conditions

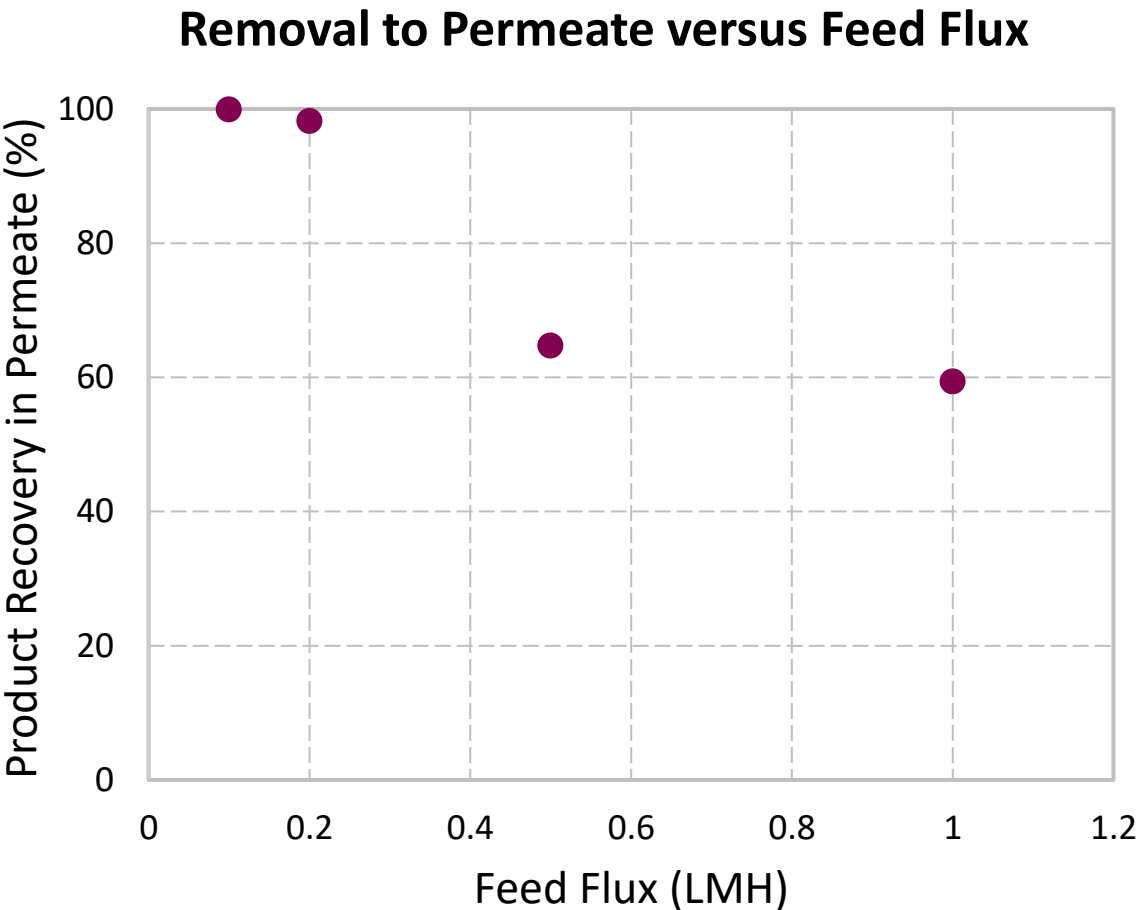
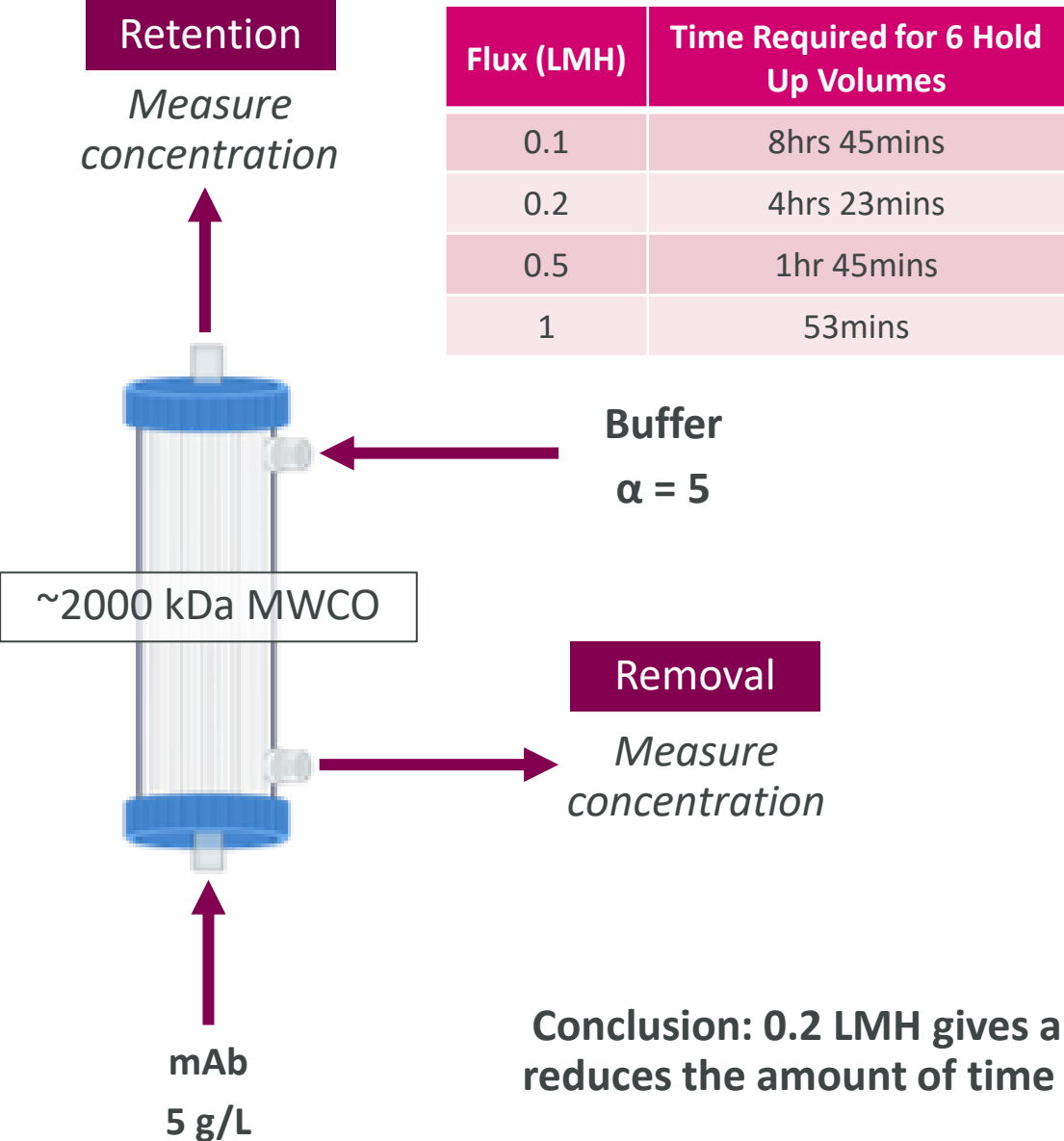
$$R_f = 1 - \frac{C_R}{C_F} = \frac{1 - \exp(-\beta A)}{\left[1 - \frac{\exp(-\beta A)}{\alpha}\right]}$$

$$\beta = \frac{\varepsilon \phi K_D D}{\delta_m q_F} \left(1 - \frac{1}{\alpha}\right)$$

Yehl, C. J., & Zydney, A. L. (2021). High performance countercurrent membrane purification for protein separations. *Journal of Membrane Science*, **633**, 119396. <https://doi.org/10.1016/j.memsci.2021.119396>



Feed Flux Evaluation



Conclusion: 0.2 LMH gives a good representation of 0.1 LMH, and reduces the amount of time required to run an experiment by half

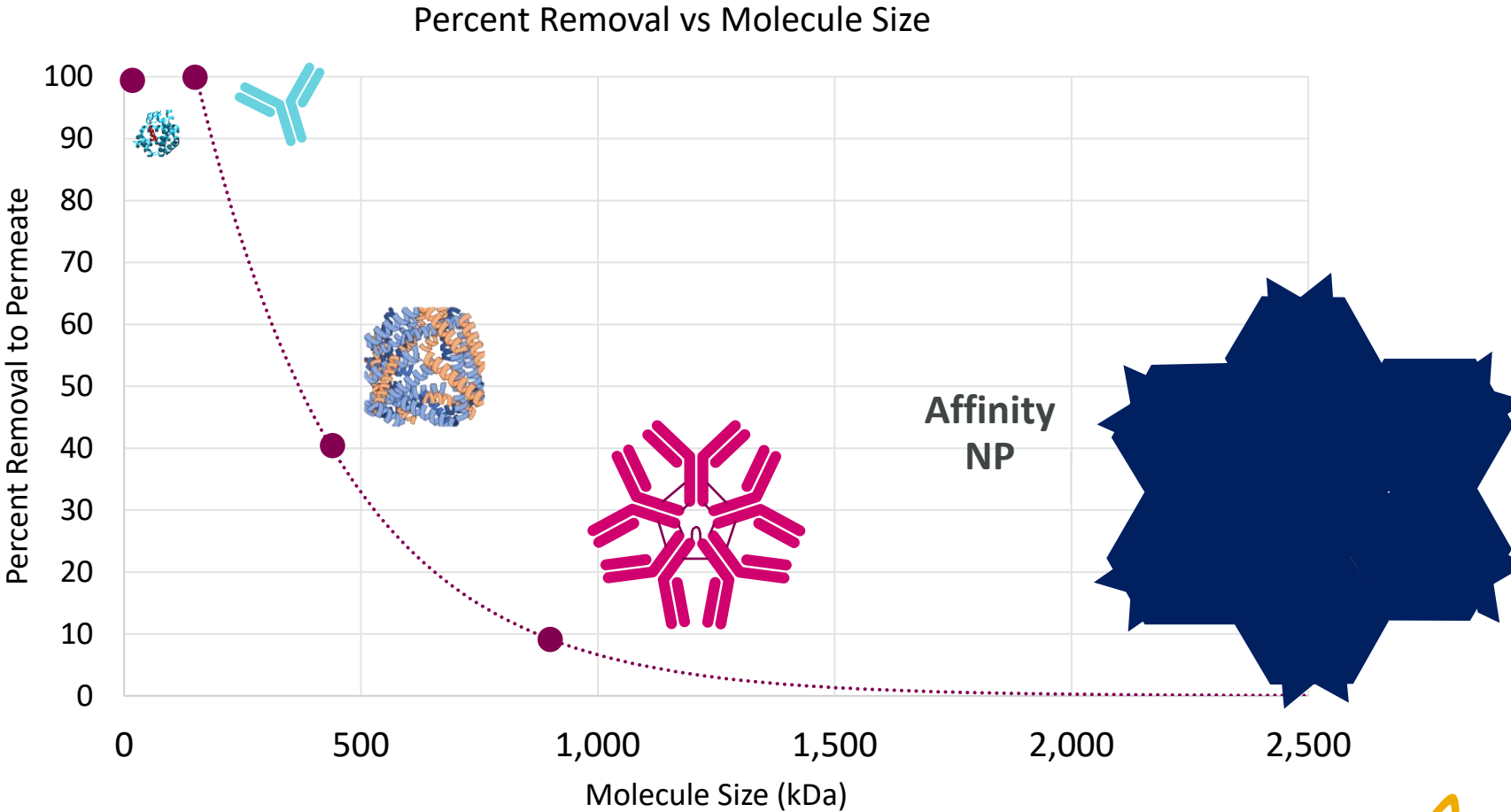


Model Proteins on Dialysis

Protein	Myoglobin	IgG	Ferritin	IgM	Affinity NP
Molecular Weight	17.6 kDa	150 kDa	440 kDa	900 kDa	2000 kDa
Diameter	3.5 nm	10 nm	12 nm	19 nm	40 nm

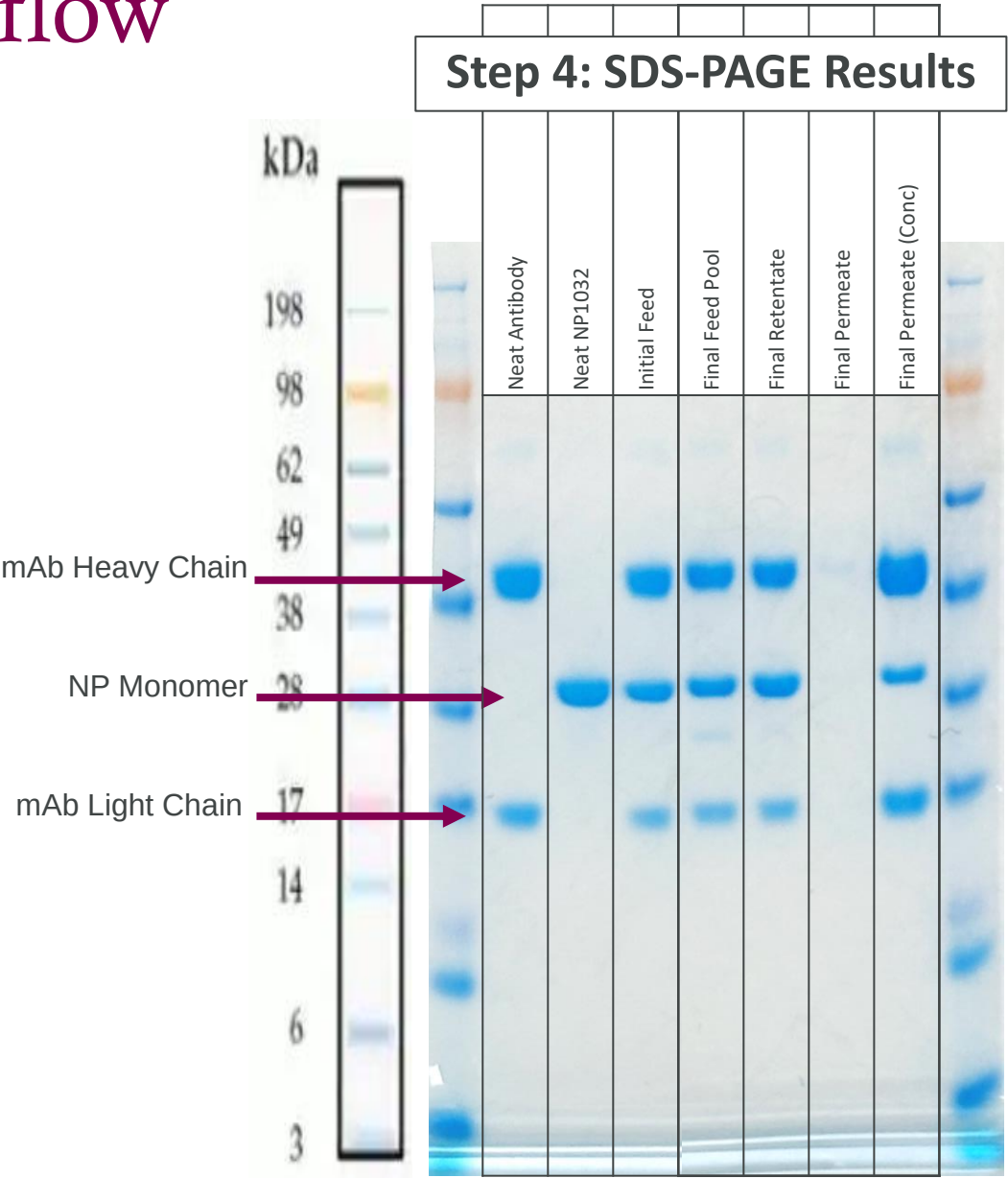
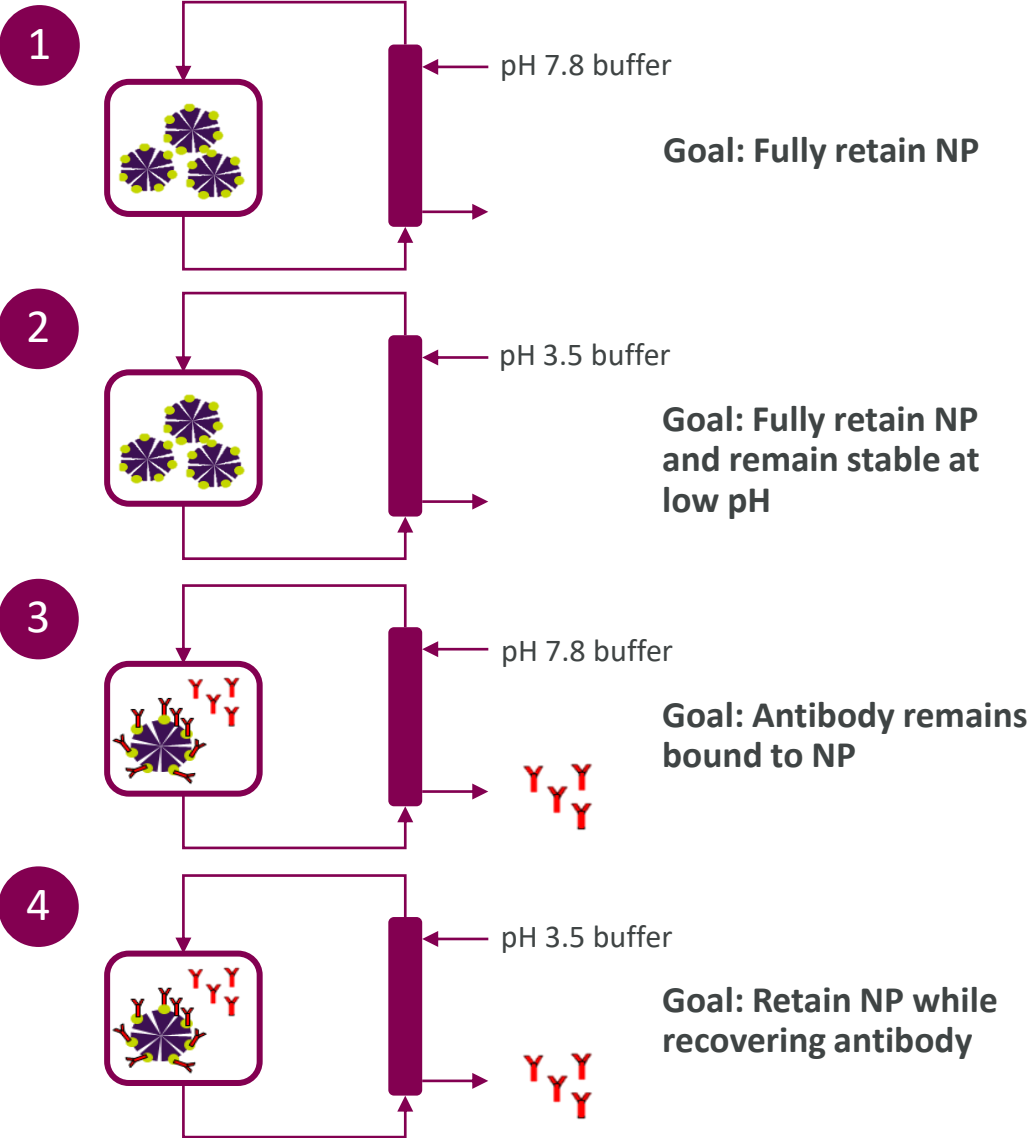
Based on the size of the nanoparticle, we would expect zero mass on the permeate. However, we see around 35% of the mass going to the permeate, which correlates to a protein size of 500 kDa and not 2000 kDa.

This shows us that the nanoparticle is not the size we think it is, or it is dynamically rearranging.



Nanoparticle Screening Workflow

Using C³ANDo



Challenges

&

Opportunities

- **Nanoparticles are likely too unstable**
 - **Dynamic nature** of the nanoparticle and mAb-NP complex in the presence of the membrane
- **Pore size distribution** of filters
- **Limited filter's** MWCO available
- **Limited analytical tools** to measure NP and complex mixtures
- TFF operation has **operational constraints** related to pressure when the complex is present (neutral pH)

- **Design nanoparticles** that are both stable at low pH and fit the current membranes
 - Stable in the presence of mAb and elution mechanism
- **Develop affinity complex** with affinity ligand small to pass through membrane while retaining mAb
- **Develop new membranes** with well characterized and tight pore size distribution
 - Small-scale, High-throughput
 - Biocompatible material
 - Representative of large-scale
- Develop analytical tools to identify and quantify nanoparticle in a mixture with mAb



Confidentiality Notice

This file is private and may contain confidential and proprietary information. If you have received this file in error, please notify us and remove it from your system and note that you must not copy, distribute or take any action in reliance on it. Any unauthorized use or disclosure of the contents of this file is not permitted and may be unlawful.

AstraZeneca PLC, 1 Francis Crick Avenue, Cambridge Biomedical Campus, Cambridge, CB2 0AA, UK
+44(0)203 749 5000
www.astrazeneca.com

